使用 LaTeX 撰寫報告

2010/12/5

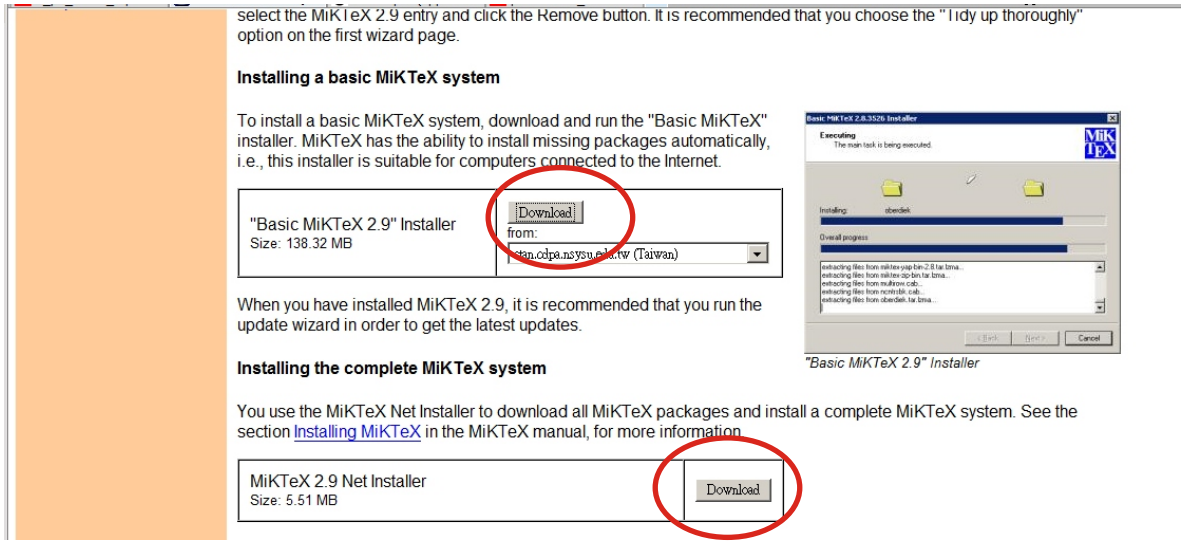
Chih-Han Lin 林致翰

clin@ltl.iams.sinica.edu.tw
or r99245002@ntu.edu.tw

安裝 TeX

Windows:

http://miktex.org/2.9/setup 下載 Miktex 2.9 安裝檔



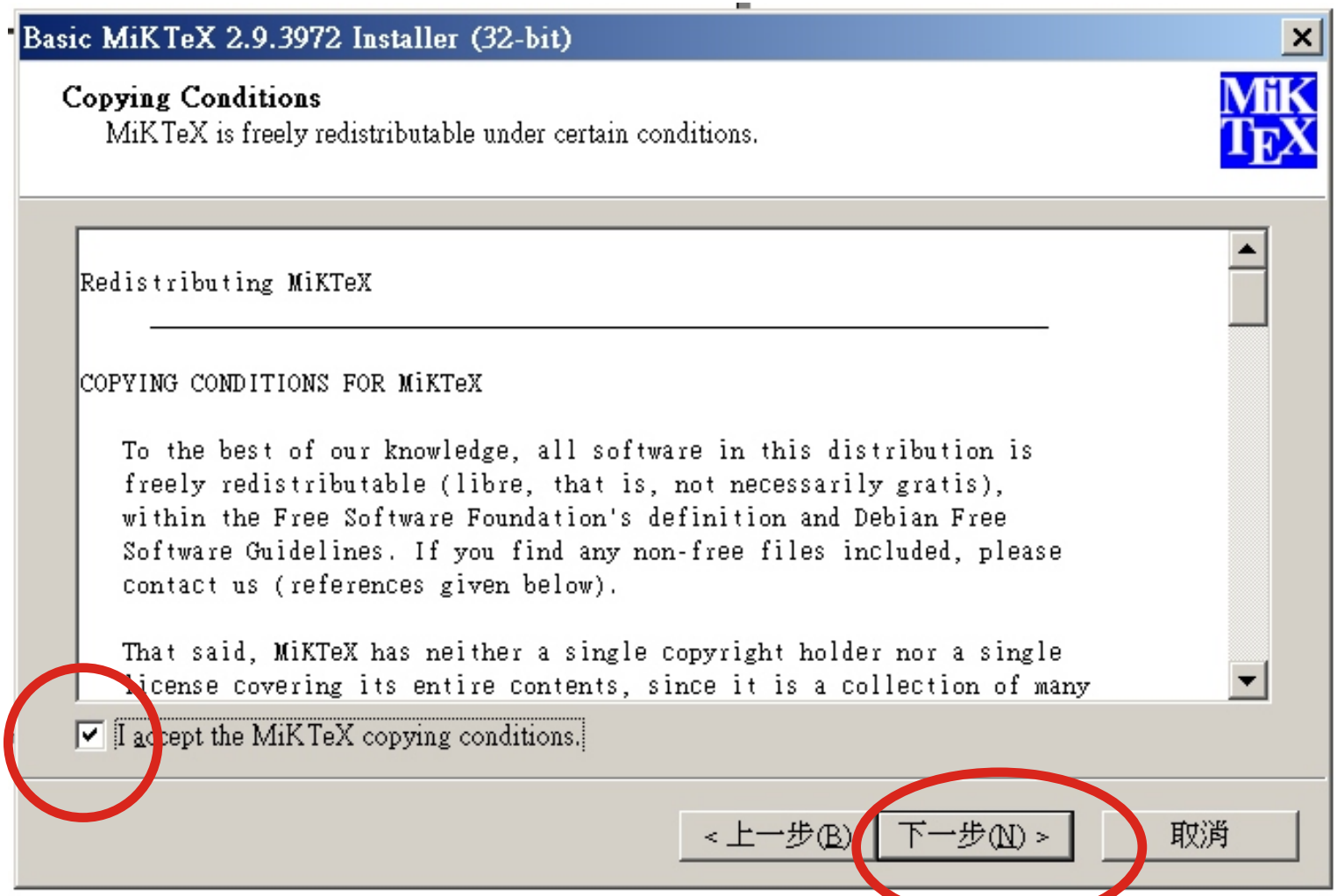
Mac OS:

請自行google

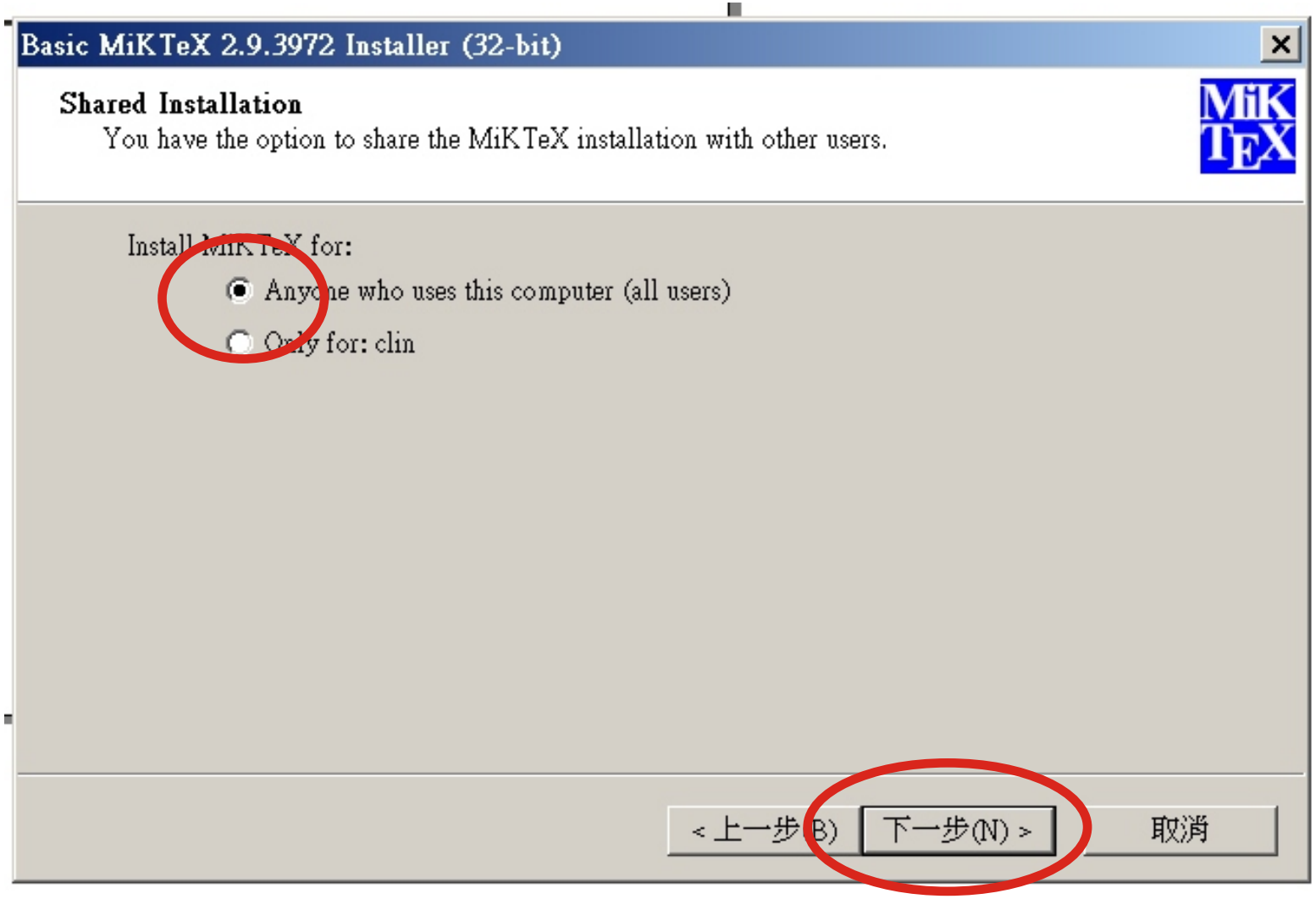
Linux:

請用該 distribution 的套件安裝程式抓相關套件

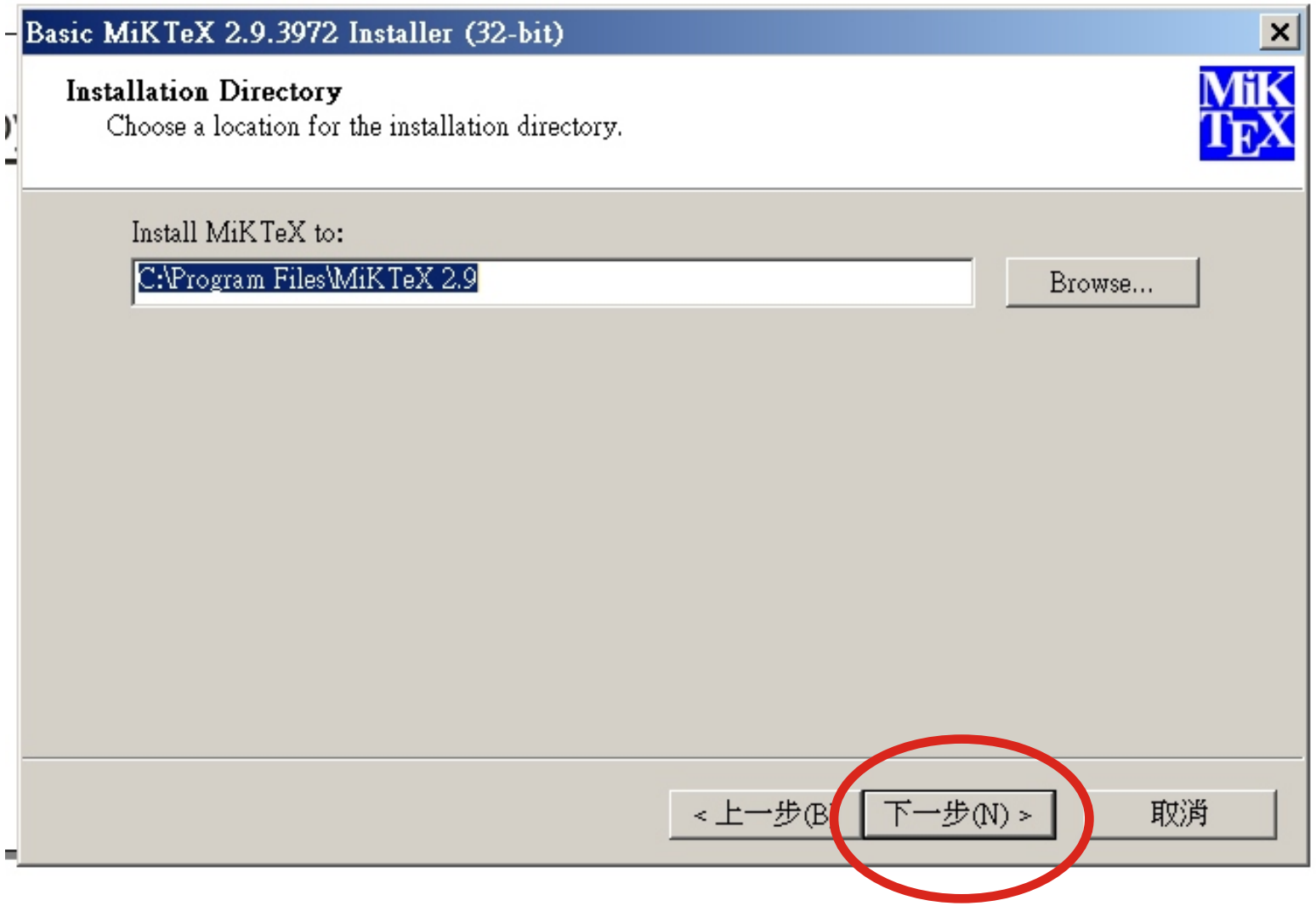
Step by Step


Basic MiKTeX 2.9.3972 Installer (32-bit)
Copying Conditions
MiKTeX is freely redistributable under certain conditions.
MiK TEX
Redistributing MiKTeX
COPYING CONDITIONS FOR MiKTeX
To the best of our knowledge, all software in this distribution is freely redistributable (libre, that is, not necessarily gratis), within the Free Software Foundation's definition and Debian Free Software Guidelines. If you find any non-free files included, please contact us (references given below).
That said, MiKTeX has neither a single copyright holder nor a single license covering its entire contents, since it is a collection of many
I accept the MiKTeX copying conditions.
< 上一步(B)
下一步(N) >
取消

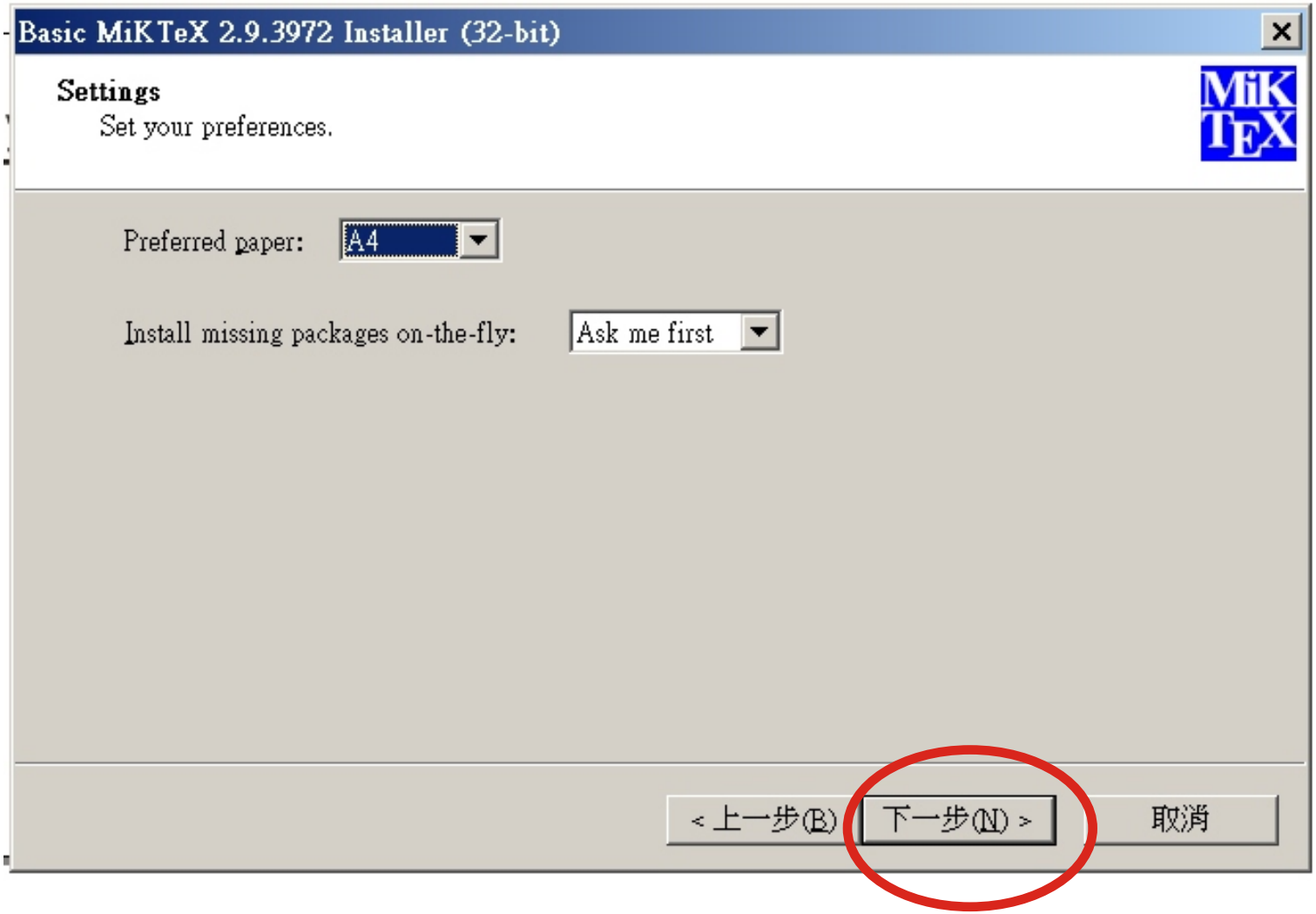
Step by Step


Basic MiKTeX 2.9.3972 Installer (32-bit)
Shared Installation
You have the option to share the MiKTeX installation with other users.
MiK TEX
Install MiKTeX for:
Anyone who uses this computer (all users)
Only for: clin
< 上一步(B)
下一步(N) >
取消

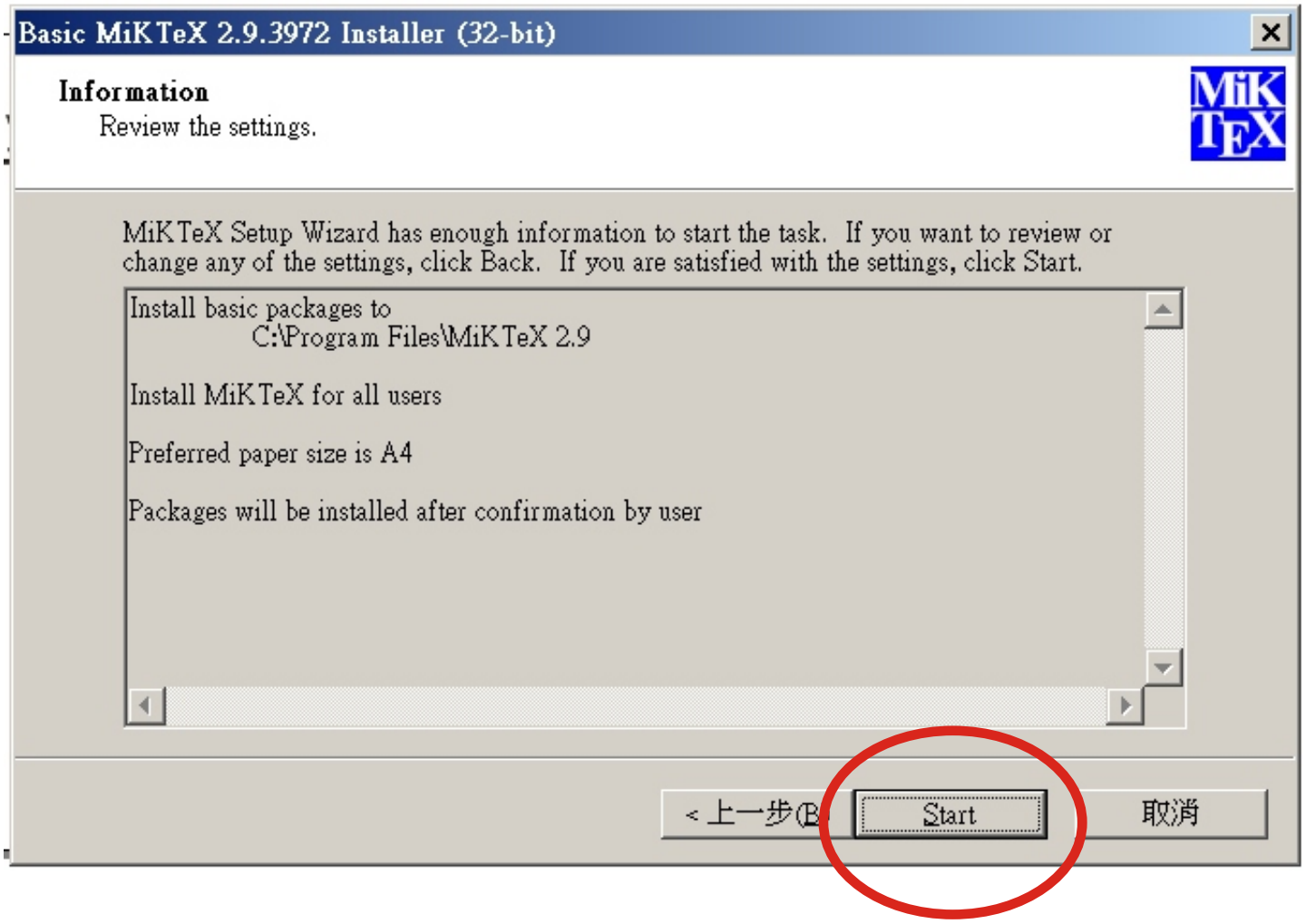
Step by Step


Basic MiKTeX 2.9.3972 Installer (32-bit)
Installation Directory
Choose a location for the installation directory.
MiK TEX
Install MiKTeX to:
C:\Program Files\MiKTeX 2.9
Browse...
< 上一步(B
下一步(N) >
取消

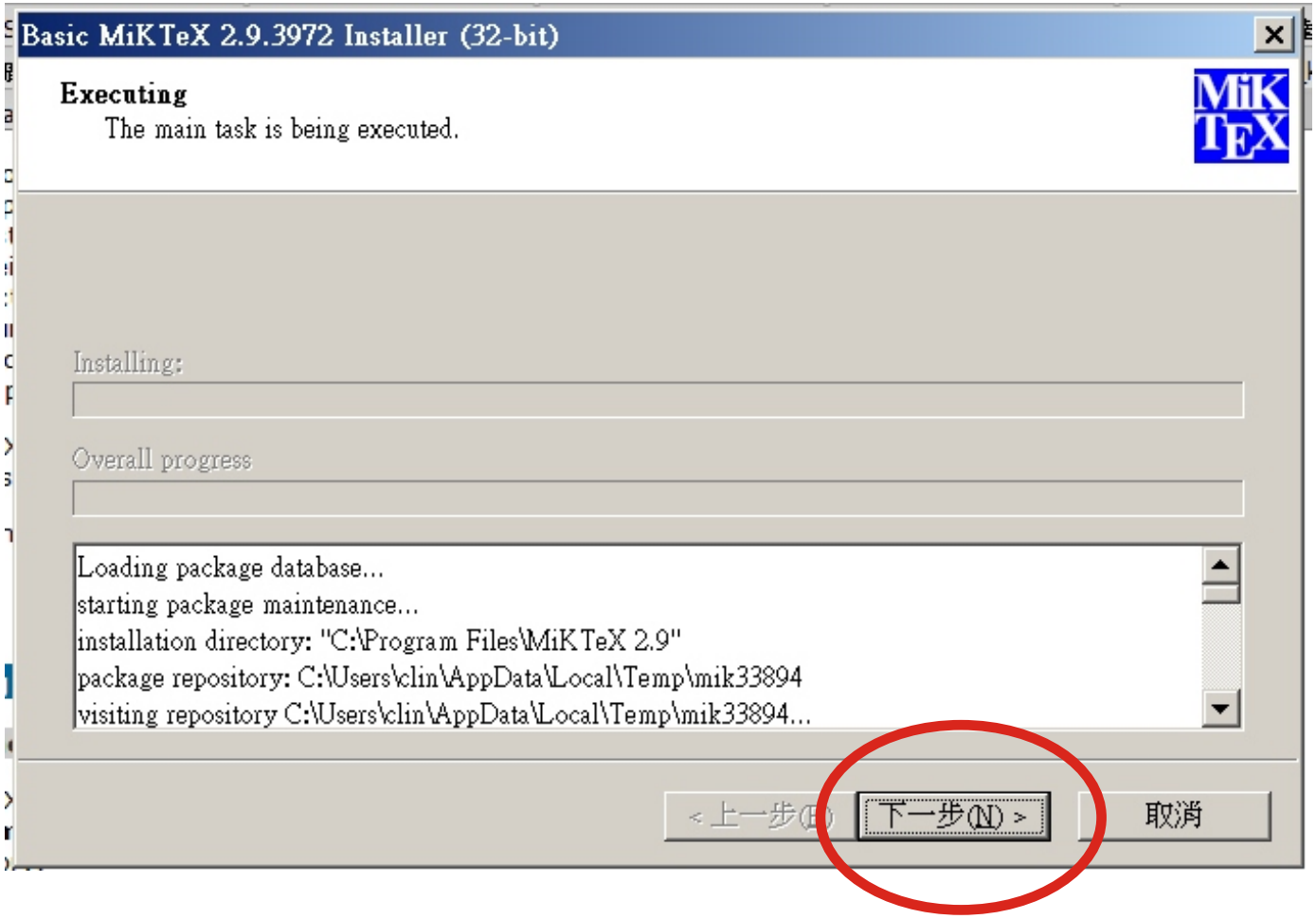
Step by Step


Basic MiKTeX 2.9.3972 Installer (32-bit)
Settings
Set your preferences.
MiK TEX
Preferred paper:
A4
Install missing packages on-the-fly:
Ask me first
< 上一步(B)
下一步(N) >
取消

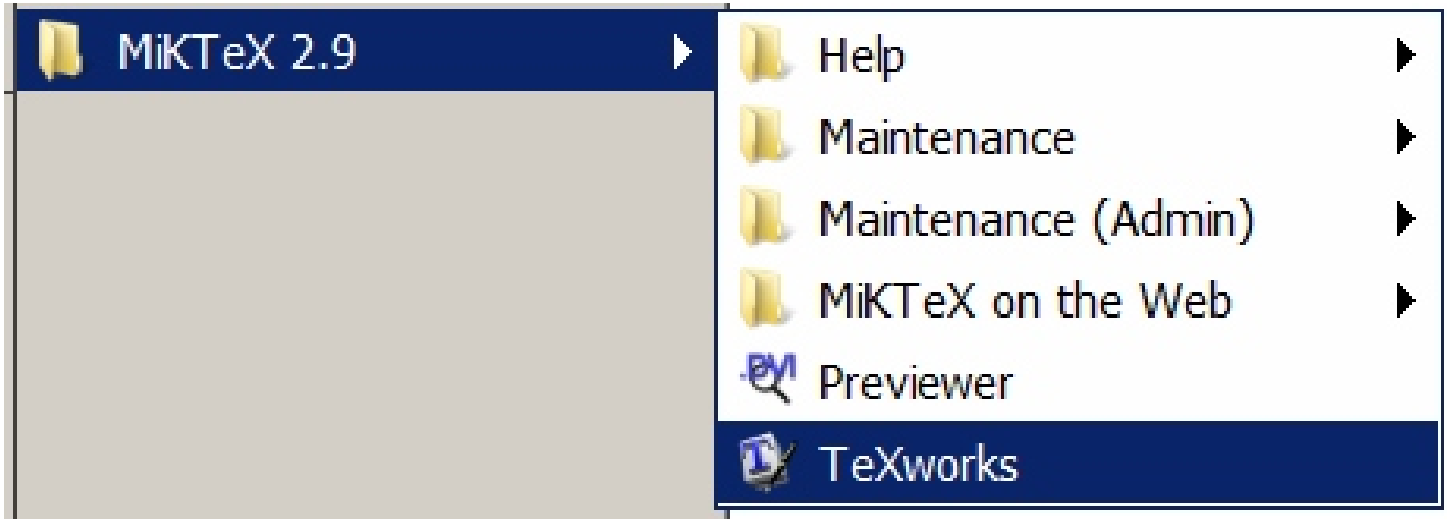
Step by Step


Basic MiKTeX 2.9.3972 Installer (32-bit)
Information
Review the settings.
MiKTeX Setup Wizard has enough information to start the task. If you want to review or change any of the settings, click Back. If you are satisfied with the settings, click Start.
Install basic packages to
C:\Program Files\MiKTeX 2.9
Install MiKTeX for all users
Preferred paper size is A4
Packages will be installed after confirmation by user
< 上一步(B
Start
取消

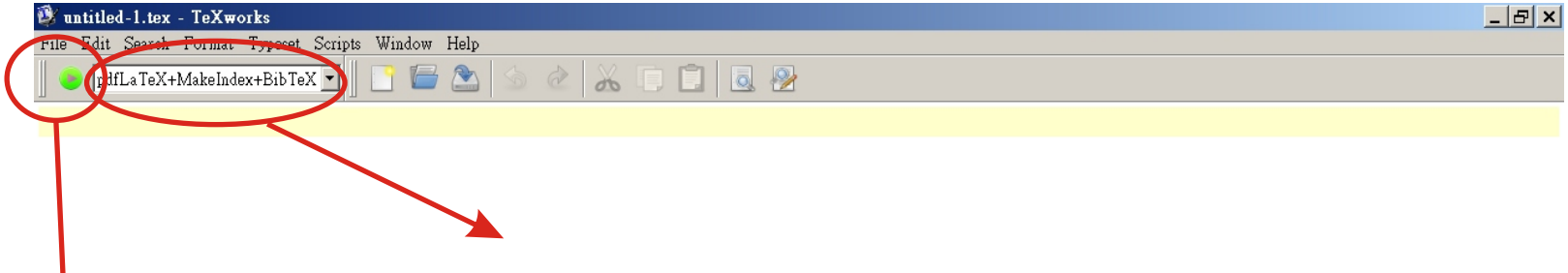
Step by Step


Basic MiKTeX 2.9.3972 Installer (32-bit)
Executing
The main task is being executed.
MiK TEX
Installing:
Overall progress
Loading package database...
starting package maintenance...
installation directory: "C:\Program Files\MiKTeX 2.9"
package repository: C:\Users\clin\AppData\Local\Temp\mik33894
visiting repository C:\Users\clin\AppData\Local\Temp\mik33894...
< 上一步(B)
下一步(N) >
取消

使用內建編輯器



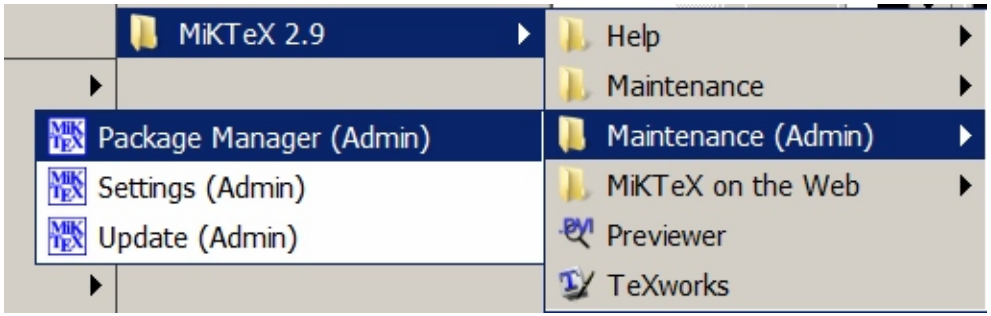
TeXworks



開始編譯

選擇編譯方式。 英文：pdflatex
中文：xelatex

安裝 package



Packege manager

點擊安裝套件

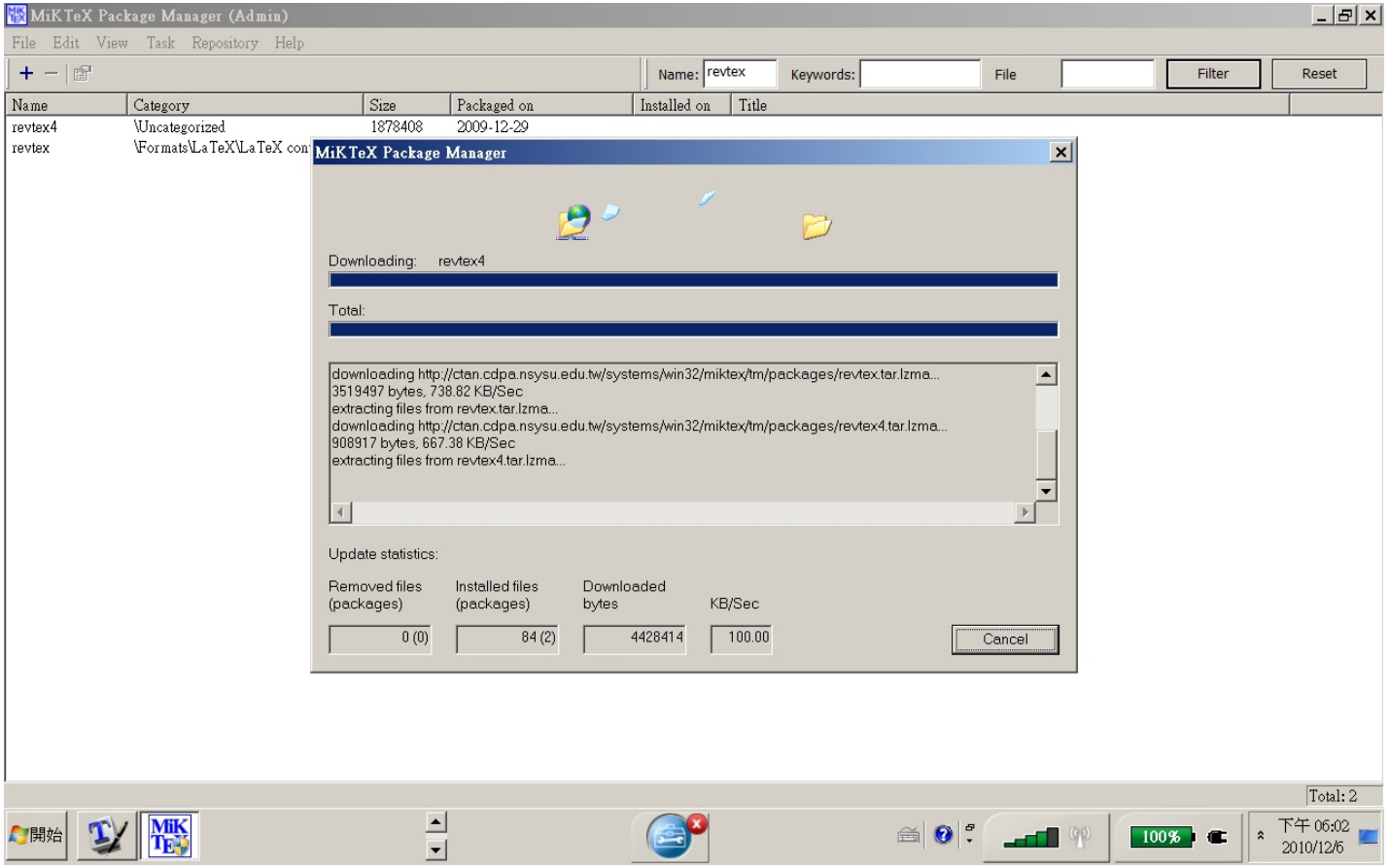
鍵入 revtex

點擊 filter

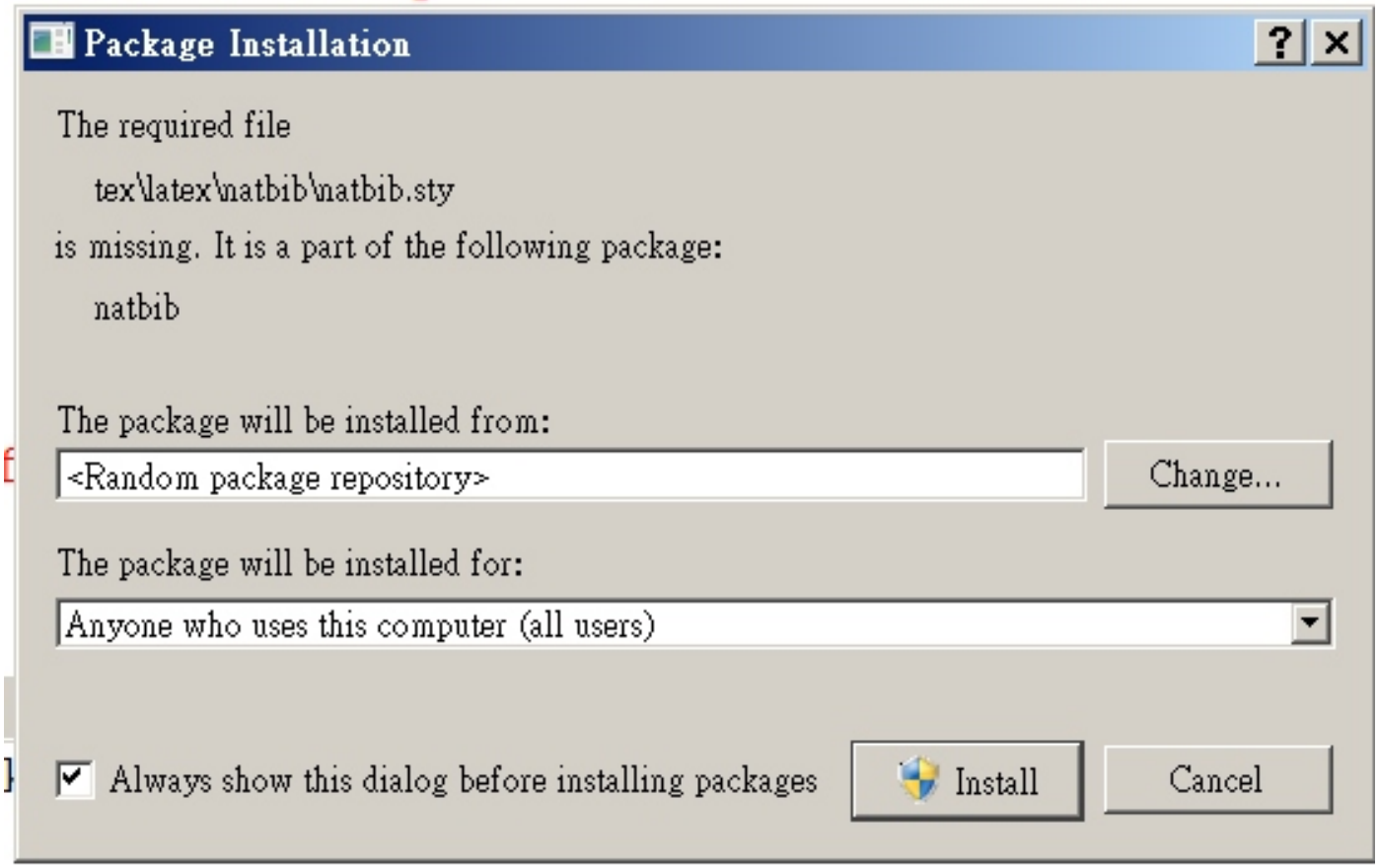


選取搜尋到的相關套件，圈選反白

安裝 package


MiKTeX Package Manager (Admin)
File Edit View Task Repository Help
Name: revtex
Keywords:
File
Filter
Reset
Name
Category
Size
Packaged on
Installed on
Title
revtex4
\Uncategorized
1878408
2009-12-29
revtex
\Formats\LaTeX\LaTeX con
MiKTeX Package Manager
Downloading: revtex4
Total:
downloading http://ctan.cdpa.nsysu.edu.tw/systems/win32/miktex/tm/packages/revtex.tar.lzma...
3519497 bytes, 738.82 KB/Sec
extracting files from revtex.tar.lzma...
downloading http://ctan.cdpa.nsysu.edu.tw/systems/win32/miktex/tm/packages/revtex4.tar.lzma...
908917 bytes, 667.38 KB/Sec
extracting files from revtex4.tar.lzma...
Update statistics:
Removed files (packages)
Installed files (packages)
Downloaded bytes
KB/Sec
0 (0)
84 (2)
4428414
100.00
Cancel
Total: 2
開始
100%
下午 06:02
2010/12/6

安裝 package



編譯文件時若缺少該文件的 package 會自動跳出 install 選單

範例（APS 期刊格式）

請從 FTP site: 課程資料 / week13 / latex / example
下載報告範本（使用 revtex4-1 APS journal 格式）

使用 Texworks 或其他搭配 Latex 快捷鍵的文字編輯器開啟
sample.tex 檔案

使用 Jabref 或其他搭配文字編輯器開啟
testbib.bib 檔案來編輯文獻資料庫

文件編譯（compile）方式：
pdfLaTeX + MakeIndex + BibTeX

Jabref 文獻管理

http://jabref.sourceforge.net/

使用 BibTeX 相容的資料格式，可輕易將編輯好文件引入 tex 文件。ISI 或各期刊網站皆可直接找到 BibTeX 格式的文章資料，不需自行建立資料庫。

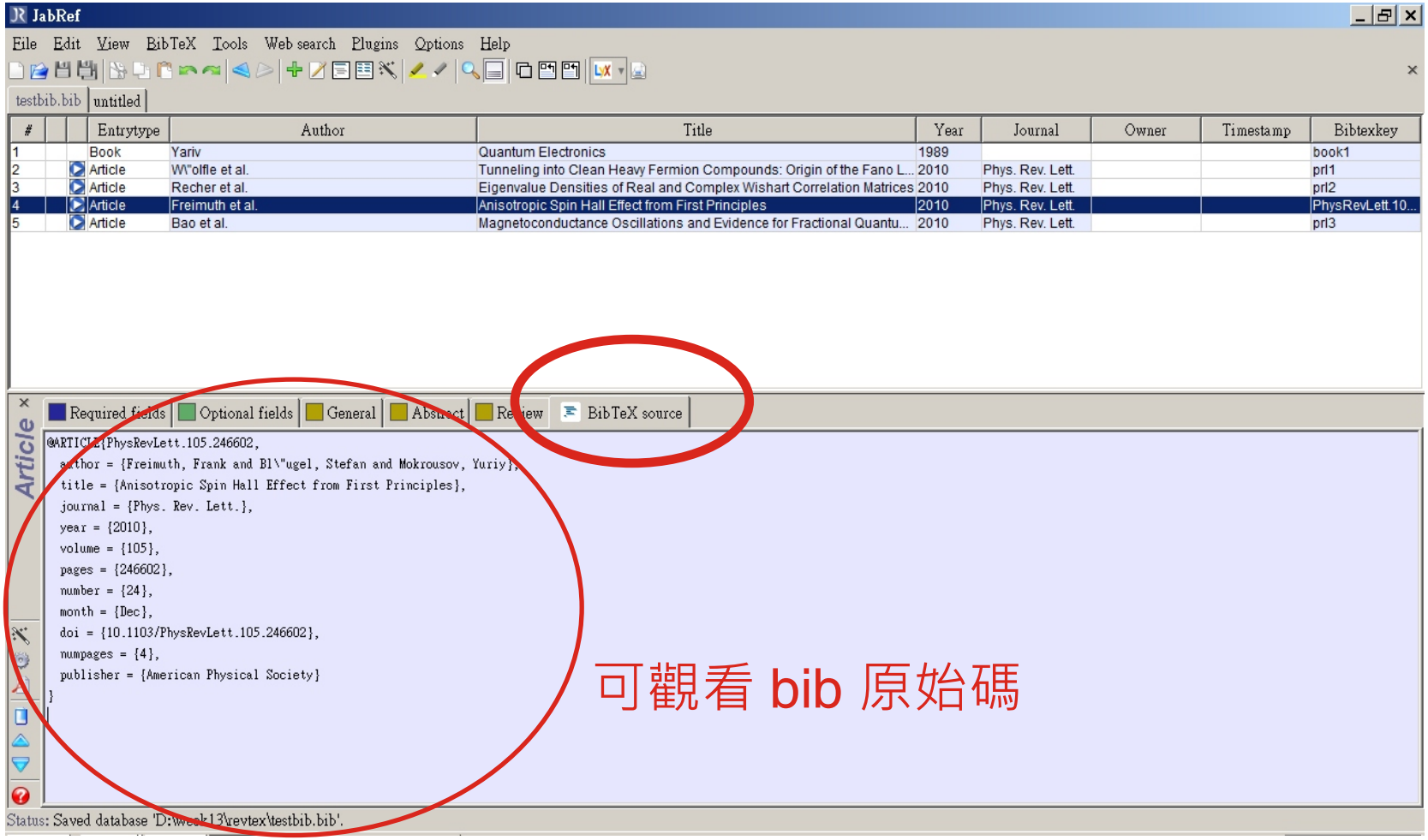
.bib 檔範例：

bibkey, 引用時的代號

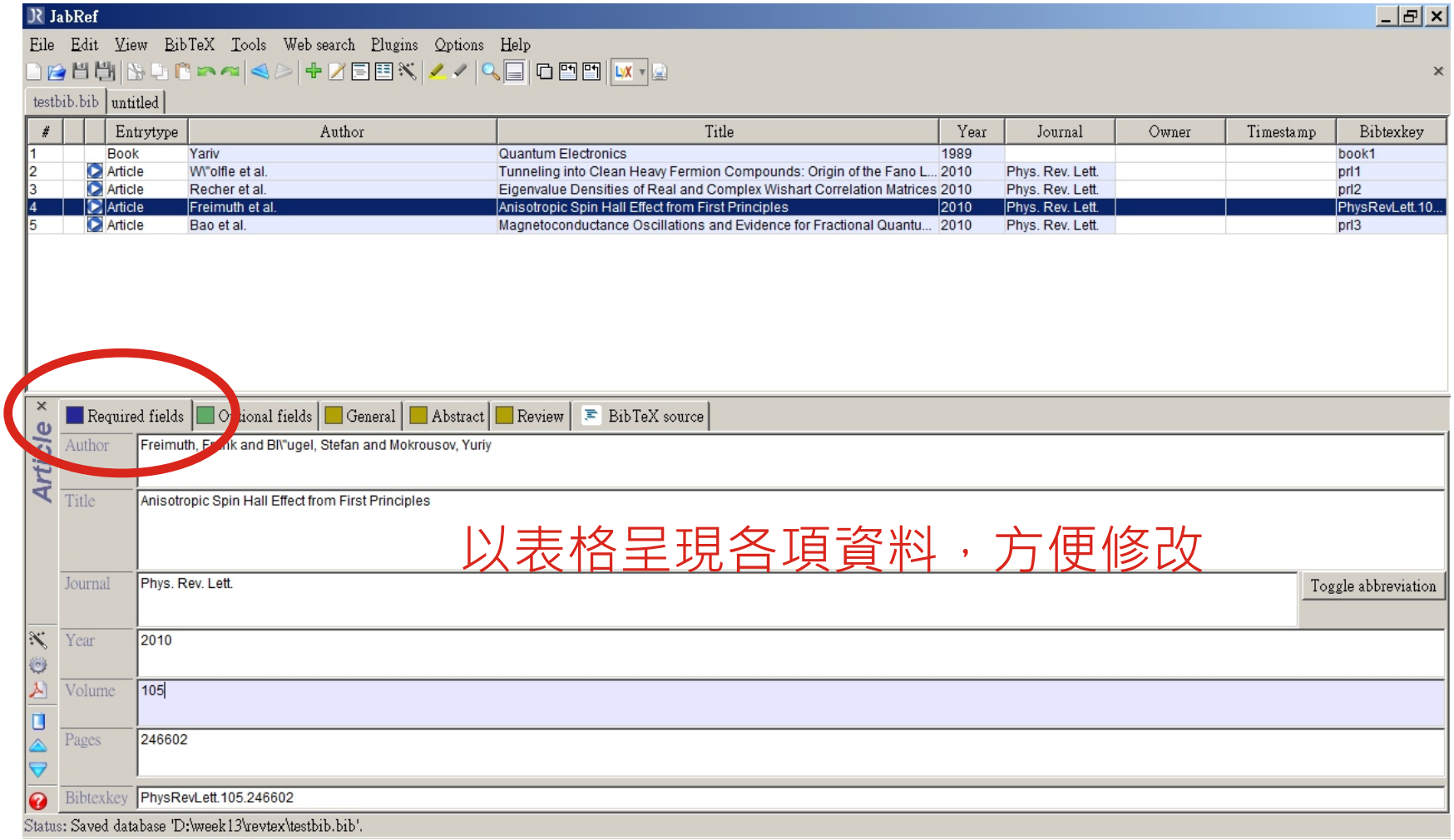
```
@Article{PhysRevLett.105.246602,
 title = {Anisotropic Spin Hall Effect from First Principles},
 author = {Freimuth, Frank  and Bl\"ugel, Stefan  and Mokrousov, Yuriy },
 journal = {Phys. Rev. Lett.},
 volume = {105},
 number = {24},
 pages = {246602},
 numpages = {4},
 year = {2010},
 month = {Dec},
 doi = {10.1103/PhysRevLett.105.246602},
 publisher = {American Physical Society}
}
```

各欄位資料

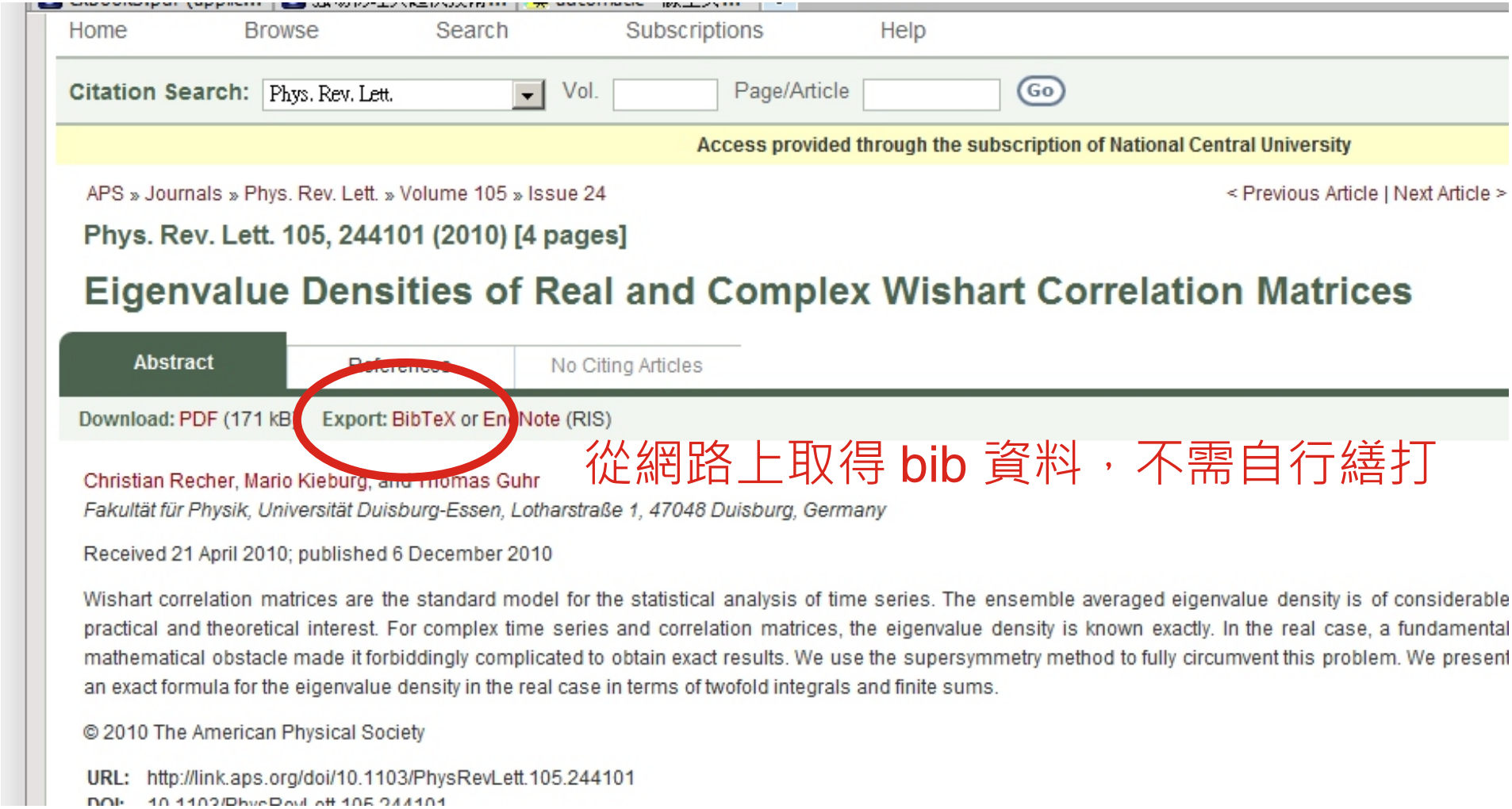
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JabRef
File Edit View BibTeX Tools Web search Plugins Options Help
testbib.bib untitled
Entrytype Author Title Year Journal Owner Timestamp Bibtexkey
1 Book Yariv Quantum Electronics 1989 book1
2 Article W\"olfle et al. Tunneling into Clean Heavy Fermion Compounds: Origin of the Fano L... 2010 Phys. Rev. Lett. prl1
3 Article Recher et al. Eigenvalue Densities of Real and Complex Wishart Correlation Matrices 2010 Phys. Rev. Lett. prl2
4 Article Freimuth et al. Anisotropic Spin Hall Effect from First Principles 2010 Phys. Rev. Lett. PhysRevLett.10...
5 Article Bao et al. Magnetoconductance Oscillations and Evidence for Fractional Quantu... 2010 Phys. Rev. Lett. prl3
Article
Required fields Optional fields General Abstract Review BibTeX source
@ARTICLE{PhysRevLett.105.246602,
author = {Freimuth, Frank and Bl\"ugel, Stefan and Mokrousov, Yuriy},
title = {Anisotropic Spin Hall Effect from First Principles},
journal = {Phys. Rev. Lett.},
year = {2010},
volume = {105},
pages = {246602},
number = {24},
month = {Dec},
doi = {10.1103/PhysRevLett.105.246602},
numpages = {4},
publisher = {American Physical Society}
}
Status: Saved database 'D:\week13\revtex\testbib.bib'.
可觀看 bib 原始碼

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testbib.bib untitled
Entrytype Author Title Year Journal Owner Timestamp Bibtexkey
1 Book Yariv Quantum Electronics 1989 book1
2 Article Wl\"olfle et al. Tunneling into Clean Heavy Fermion Compounds: Origin of the Fano L... 2010 Phys. Rev. Lett. prl1
3 Article Recher et al. Eigenvalue Densities of Real and Complex Wishart Correlation Matrices 2010 Phys. Rev. Lett. prl2
4 Article Freimuth et al. Anisotropic Spin Hall Effect from First Principles 2010 Phys. Rev. Lett. PhysRevLett.10...
5 Article Bao et al. Magnetoconductance Oscillations and Evidence for Fractional Quantu... 2010 Phys. Rev. Lett. prl3
Required fields
General
Abstract
Review
BibTeX source
Author
Freimuth, and Bl\"ugel, Stefan and Mokrousov, Yuriy
Title
Anisotropic Spin Hall Effect from First Principles
以表格呈現各項資料，方便修改
Journal
Phys. Rev. Lett.
Toggle abbreviation
Year
2010
Volume
105
Pages
246602
Bibtexkey
PhysRevLett.105.246602
Status: Saved database 'D:\week13\revtex\testbib.bib'.

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Page/Article
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APS » Journals » Phys. Rev. Lett. » Volume 105 » Issue 24
< Previous Article | Next Article >
Phys. Rev. Lett. 105, 244101 (2010) [4 pages]
Eigenvalue Densities of Real and Complex Wishart Correlation Matrices
Abstract
No Citing Articles
Download: PDF (171 kB)
Export: BibTeX or EndNote (RIS)
從網路上取得 bib 資料，不需自行繕打
Christian Recher, Mario Kieburg, and Thomas Guhr
Fakultät für Physik, Universität Duisburg-Essen, Lotharstraße 1, 47048 Duisburg, Germany
Received 21 April 2010; published 6 December 2010
Wishart correlation matrices are the standard model for the statistical analysis of time series. The ensemble averaged eigenvalue density is of considerable practical and theoretical interest. For complex time series and correlation matrices, the eigenvalue density is known exactly. In the real case, a fundamental mathematical obstacle made it forbiddingly complicated to obtain exact results. We use the supersymmetry method to fully circumvent this problem. We present an exact formula for the eigenvalue density in the real case in terms of twofold integrals and finite sums.
© 2010 The American Physical Society
URL: http://link.aps.org/doi/10.1103/PhysRevLett.105.244101

引用圖片

圖片最好放在與引用的 tex 文件同目錄底下，使用相對路徑引入

```
\usepackage{graphicx}
```

preamble 必須加入 graphicx 巨集

```
\begin{figure}
\centering
\includegraphics[width=0.4\textwidth]{test-pic.pdf}
\caption{This is test-pic picture}
\label{test-pic}
\end{figure}
```

置中

圖片檔案連結

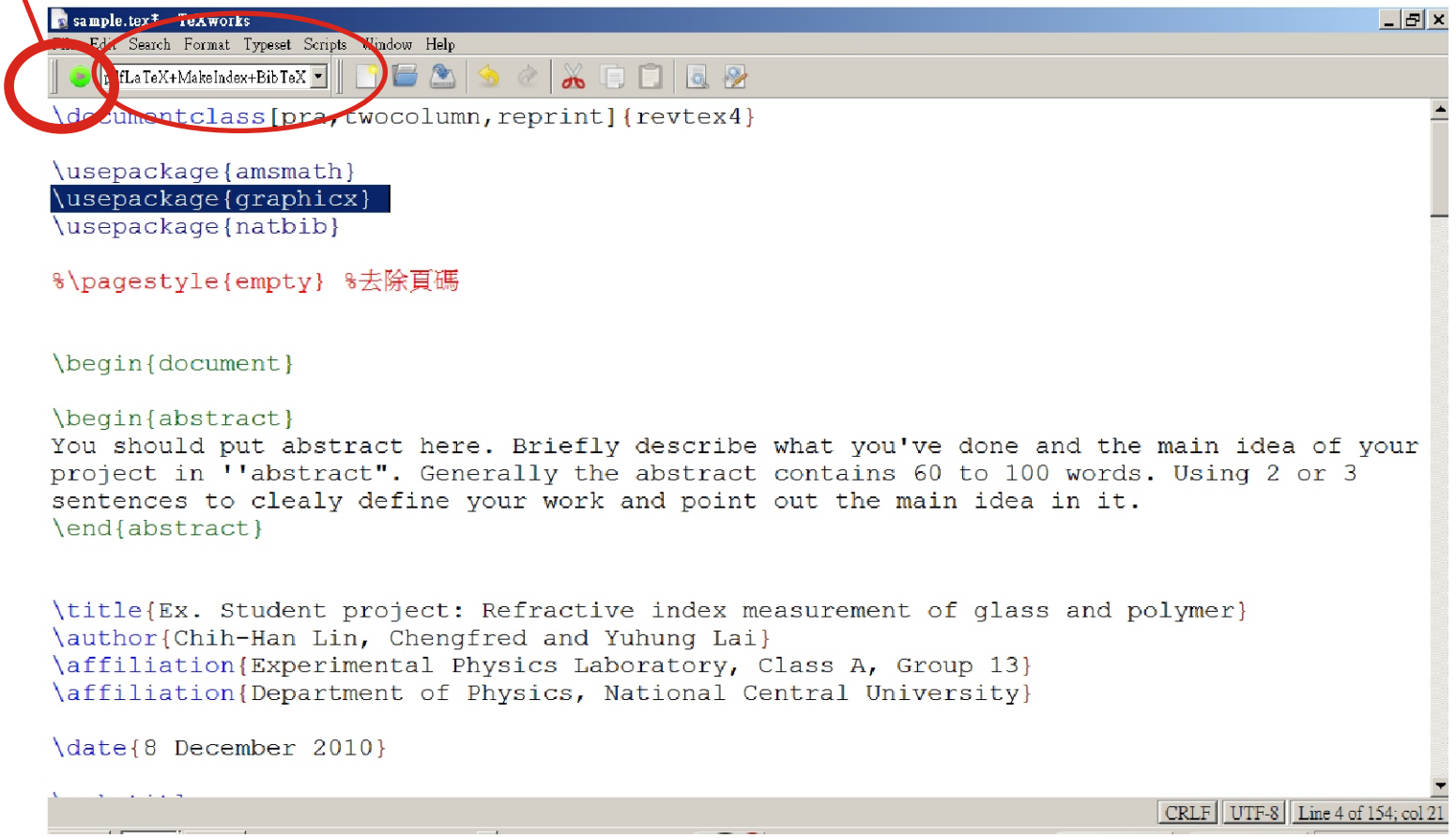
圖片標題

設定寬度，由於文章為 two-column,
這邊建議使用 0.4 倍文寬

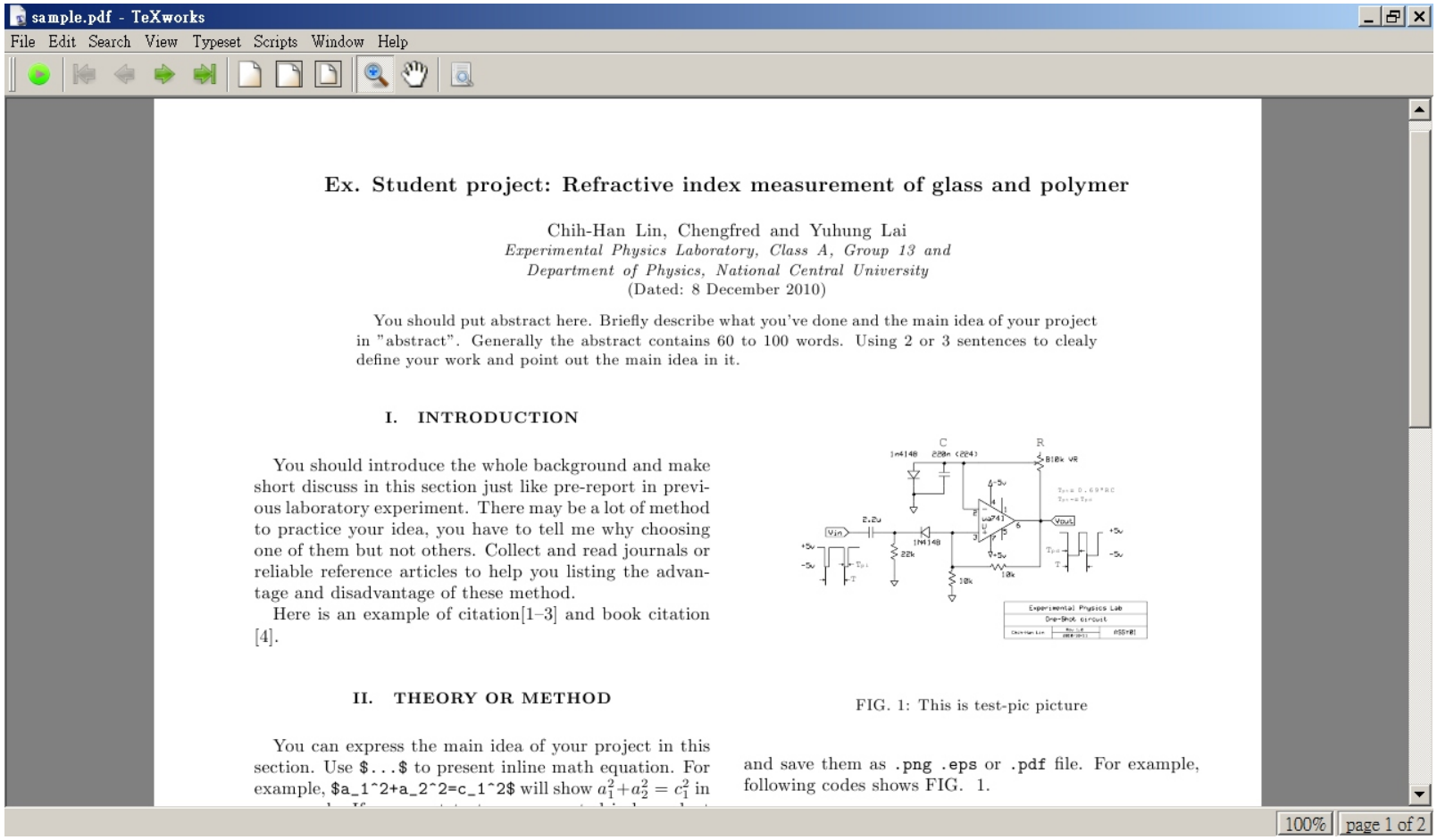
範例

編譯文件

選 pdfLaTeX + MakeIndex + BibTeX



範例



LaTeX 輸出結果為 pdf 檔

hint

一個文件的所有檔案（.tex, .bib, picture）獨立放在一個資料夾中。
文件編譯過程會產生許多中介檔案（.aux, .log 等等）

文件不要放在有中文檔名的目錄下。

檔案編輯出錯時，在下方對話框輸入 r 按 enter 可略過錯誤繼續編譯

太複雜的表格可以先用繪圖軟體繪製當成圖片加入文檔。
但請勿轉換點陣圖！務必使用 .eps 或 .pdf 檔